



Towards Water Sensitivity: A Critical Review of Urban Water Management Strategies

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Abstract

Rapid urbanization and climate change pose unprecedented challenges to urban water systems, necessitating a holistic and integrated approach to sustainable water resource management. Few studies have focused on enabling water-sensitive development by co-creating a Water Sensitive Cities (WSC) framework with fit-for-purpose guidelines, addressing site-specific issues, knowledge, practices, and proposing new interventions and policies. This review summarizes the evolution of urban water management practices, highlighting the WSC framework as a blueprint for achieving resilient, equitable, and sustainable urban water systems. An initial search in Google Scholar and Web of Science using relevant keywords yielded 831 articles. After filtering for topic relevance, 741 articles were selected for this review. The reviewed studies cover a wide geographical range, including Global North (United Kingdom, Germany, France, Singapore, USA, and Australia) and Global South regions (India, China, Sub-Saharan Africa, and parts of Latin America). Socio-ecological factors are considered to guide the selection of appropriate technology and infrastructure. The diverse experiences and challenges in constructing WSCs are analyzed through case studies from different geographies around the globe. Additionally, the review advocates for decentralized solutions, green infrastructure, and active community engagement to ensure successful WSC implementation. The review concludes with recommendations for future research and practical applications in urban water management.

Keywords Urbanization · Water sensitive city · Integrated water management · Scenario-based analysis · Green infrastructure

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1 Introduction

The integrative and fit-for-purpose water-sensitive design framework has become an essential concern for rapidly growing liveable cities around the world (Lechner et al. 2020; Bertrand-Krajewski, 2021). Rapidly growing cities are defined as urban centres experiencing rapid urbanization (Litman 2016; Abraha et al. 2022). This is characterized by high-velocity demographic and spatial expansion that significantly outpaces the planning, financing, and implementation of formal water infrastructure (Bahri 2012; UN-Habitat 2022). This phenomenon leads to unsustainable patterns (Cavoli 2021), including (i) rapid, unregulated land-use change and urban sprawl; (ii) the proliferation of informal settlements and peri-urban areas disconnected from formal governance (Nassary et al. 2022); and (iii) the emergence of persistent infrastructure and governance gaps (Bahri 2012). It is this rapidity of change that creates a widening gap between water demand and supply, and institutional fragmentation, resulting in growing water risk challenges (Abraha et al. 2022). These cities face urgent and uncertain challenges with their urban water systems, for which a formulation is required to mitigate and cope with the complex short- and long-term challenges (Rijke et al. 2012). For example, challenges such as flash flood or drought in urban areas have become very frequent and associated mitigation measures have been evaluated in studies through engineering solutions (e.g., rain gardens, bioswales, and green roofs etc.), land use planning (e.g., flood plain zoning, preservation of wetland, and permeable pavement etc.), and community engagement (citizen awareness regarding water sensitivity). An integrative and socio-ecological approach is required to facilitate sustainability and resilience of the urban water system in the long run (Frantzeskaki et al. 2018). An integrative approach refers to managing urban water systems in a holistic, cross-disciplinary, and cross-sectoral manner recognizing the interdependencies among water, the environment, urban planning, infrastructure, governance, and society (Hejnowicz et al. 2022). In addition, the socio-ecological approach views urban water systems as embedded within complex, dynamic interactions between humans and ecosystems (Madrid-López and Giampietro 2015).

In recent years, integrated urban water management (IUWM) has become increasingly important in the context of global urbanization and climate change. Some of the successful strategies published in literature are low-impact development in China (Xu et al. 2017), sustainable urban drainage systems in UK (Ellis and Lundy 2016), water-sensitive urban design in Australia (Liu et al. 2016), the active, beautiful, and clean waters program in Singapore (Lim and Lu 2016), and the sponge city initiative in China, (Wang et al. 2018). The technological solutions for water management range from small-scale household solutions, such as rainwater harvesting, to centralized solutions, including desalination and large dams (Makropoulos and Butler 2010). Water Sensitive City (WSC) is a popular concept and is based on three pillars: (i) cities as a water supply catchment, (ii) cities providing ecosystem services in relation to the built and natural environment, and (iii) making the community of the city water-sensitive (Wong and Brown 2009). In recent years, many definitions of WSC have expanded beyond sustainability to encompass intra-generational justice and equity. As WSC aims to ensure “best-practice urban water management by considering the supply security, public health protection, flood protection, waterway health protection, amenity and recreation, greenhouse neutrality, economic vitality, intra and inter-generational equity; and demonstrable long-term environmental sustainability” (Rogers et al. 2020). The evolution of cities progresses through six distinct stages, with WSC being the ultimate stage (Bichai

and Cabrera Flamini 2018). A benchmarking approach is needed to evaluate the stage of evolution, and the fit-for-purpose solutions need flexible options of technical and management solutions at different scales.

Urban areas are adapting to overcome the difficulties and make use of opportunities arising from climate change and other changes, which are both uncertain and complex. The secondary cities of the global south are adapting to deficits in infrastructure and basic services. The integration of adaptation responses is gaining prominence following the Paris 2015 accord, as governments at various levels worldwide have begun planning and implementing adaptation measures. Additionally, understanding and integrating adaptation responses are crucial for achieving the Sustainable Development Goals (SDGs) (Fund 2015). The WSC concept can be applied by systematically categorizing literature based on its relevance to the three pillars of WSC: (i) cities as water supply catchments, (ii) provision of ecosystem services, and (iii) fostering water-sensitive communities (Wong and Brown 2009).

Only a limited number of studies have primarily focused on enabling water-sensitive development by co-creating a framework for WSC and developing fit-for-purpose guidelines (Figs. 1 and 2), accounting for site-specific issues, knowledge, and practices, while proposing new interventions and policy guidelines (Brown et al. 2009). Fit-for-purpose solutions require flexibility across multiple scales, such as rainwater harvesting (household scale), green corridors (neighbourhood scale), and integrated drainage and wastewater planning (city-wide scale) (Makropoulos and Butler 2010). The solutions require flexibility because every scale—local, regional, or national—presents unique environmental, social, technical, and managerial contexts (Rijke et al. 2012). This review's novelty lies in its comprehensive approach to WSC by co-creating a “fit-for-purpose” framework with adaptable guidelines. It uniquely integrates socio-ecological factors for technology selection, analyzes diverse global case studies to inform context-specific solutions, and strongly advocates for decentralized solutions, green infrastructure, and active community engagement as essential for sustainable urban water management in rapidly urbanizing regions. It examines global approaches like sustainable urban drainage systems (SUDS), low impact development (LID), and integrated urban water management (IUWM) and highlights the essential role of interdisciplinary collaboration and community engagement in developing holistic strategies. Community engagement is integrated into WSC approaches through participatory planning, citizen awareness campaigns, and co-design of decentralized solutions (Biome Environmental Trust, 2017). For example, participatory aquifer mapping in Bangalore demonstrated that local knowledge can be harnessed to manage groundwater sustainably. Thus, the model integrates both the challenges and opportunities of community involvement in rapidly urbanizing contexts.



Fig. 1 Features of water sensitive city (watersensitivecities.org.au/what-is-a-water-sensitive-city/)

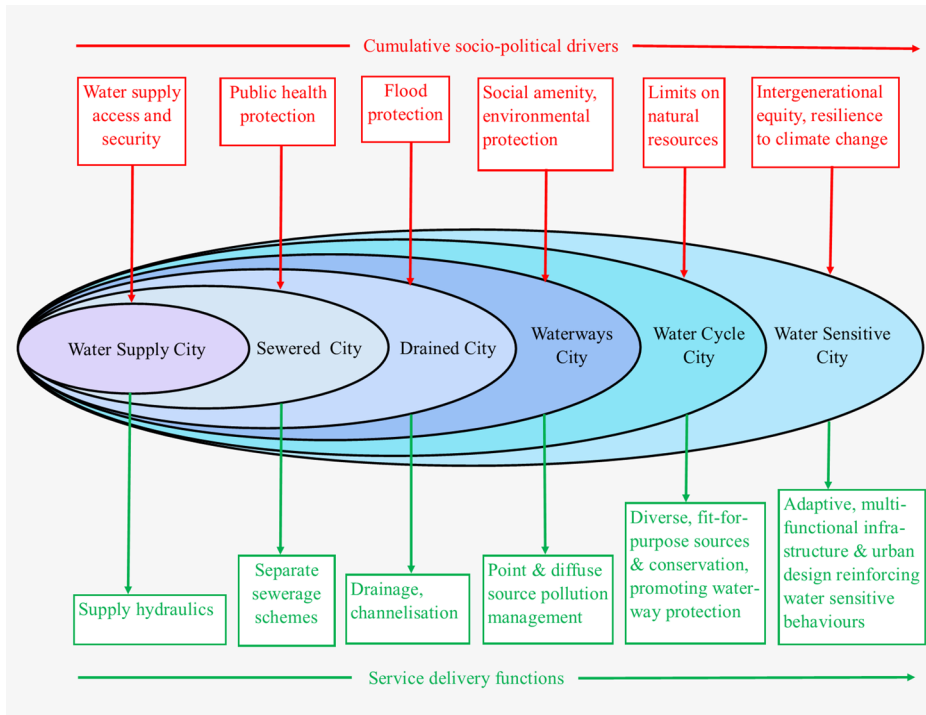


Fig. 2 Urban Water Transitions Framework (Reproduced from Brown et al. 2009)

The structure of this review paper is designed to provide a comprehensive state-of-the-art and critical understanding of the concepts, methodologies, and practices involved in sustainable urban water management, particularly within the context of rapidly growing cities.

2 Methodology

The methodology section outlines the systematic approach, including the search strategy, study selection criteria, data extraction, and data synthesis.

2.1 Search Strategy

A comprehensive literature search was conducted using Web of Science and Google Scholar, for their extensive databases and broad disciplinary reach. This search included publications in English from 1980 to 2024, capturing the historical evolution of WSC and related concepts. The search strategy involved specific terms like “Water-sensitive city” OR “Integrated urban water management” OR “Sustainable urban water management”, allowing for a comprehensive range of studies that focus on similar water management systems and approaches. Additionally, a manual review of the reference lists from selected articles was conducted to identify other relevant studies.

2.2 Inclusion and Exclusion Criteria

Selected literature was assessed based on criteria for relevance, source credibility, and methodological robustness. Studies were included in this review if they met the following criteria: (a) The study was published in a peer-reviewed journal. (b) The study focused on the topic of integrated urban water management and/or sustainable urban water management. (c) The study provided substantial information and data relevant to the objectives of this review. (d) The study was published within the specified time frame. Studies were excluded if they: (e) were not published in English. (f) Were conference abstracts, posters, or unpublished manuscripts. (g) Were not relevant to the topic of interest. (h) Did not provide sufficient data or information to meet the review's objectives. To minimize bias, this review considered diverse geographical regions and both theoretical and applied research, aiming to balance practical case studies with conceptual frameworks. Through this careful selection process, the review aimed to provide a holistic and accurate representation of current knowledge and research gaps within urban water management.

2.3 Study Selection criteria

The inclusion and exclusion criteria for selecting research papers yielded 972 records, which were screened for language, resulting in 910 records. These records were further screened for full-text availability, reducing the number to 831. Subsequently, the records were screened for keyword availability, narrowing the list down to 778, which were then assessed for topic relevance. Three independent authors conducted the initial screening of titles and abstracts. Any discrepancies in study selection were resolved through discussion and consensus. The full texts of selected studies were then assessed against the inclusion and exclusion criteria. In cases of persistent disagreement, a fourth author was consulted. A flowchart is shown in Fig. 3.

2.4 Data Synthesis

After identifying the research papers for review, their content were synthesized in a structured format, categorizing them by "Paper title," "Location," "Purpose," "Parameters considered," "Type of demand considered," "Analysis/Methodology/Data," "Remarks," and "Reference" (Table S1). Following this, a narrative synthesis approach was employed to summarize the overall findings. Key themes and patterns were identified, and the results were systematically presented to provide a comprehensive understanding of the research landscape.

3 Different Terminologies

Over time, various terminologies have emerged to describe different components and processes. Some key terminologies are elaborated below:

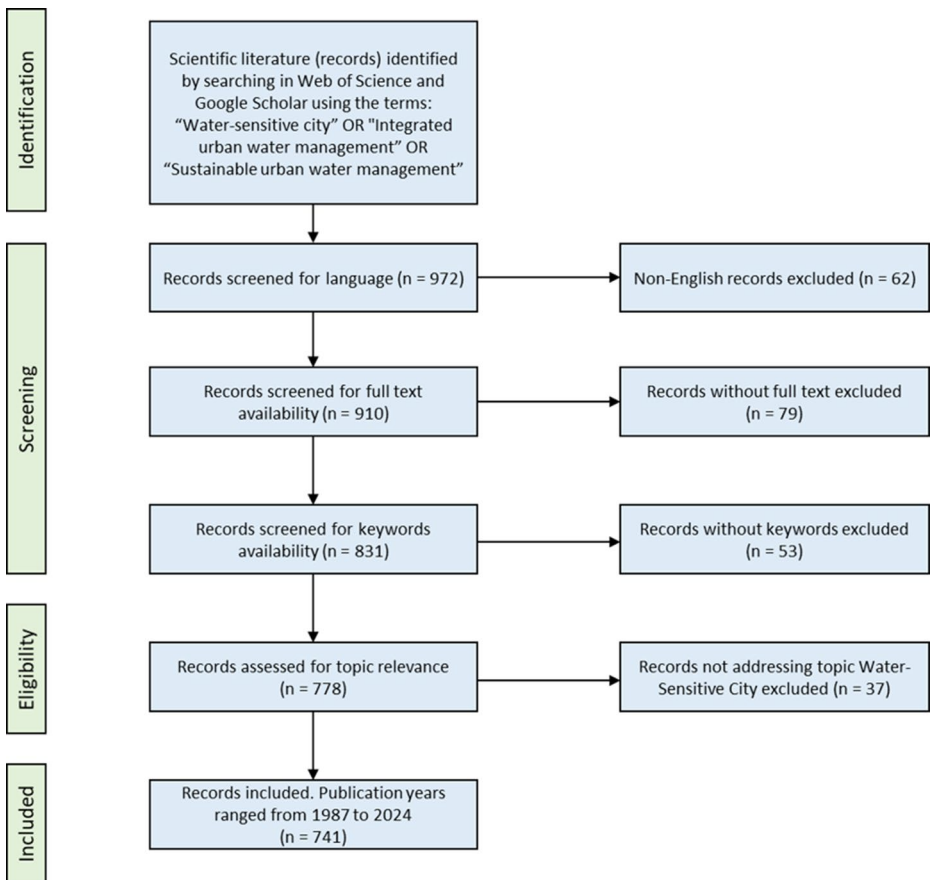


Fig. 3 Flowchart illustrating the literature selection process for the present study

3.1 Sustainable Urban Water Management (SUWM)

The concept of a water-sensitive city (WSC) integrates the principles of Sustainable Urban Water Management (SUWM) with essential urban water services (Bichai and Cabrera Flamini 2018). A WSC embodies SUWM values, including environmental protection, equity, rehabilitation, and sustainability, while also ensuring critical water services like supply security, flood control, and public health (Wong and Brown 2009; Ferguson et al. 2013).

Figure 4 presents a global framework for sustainable water resource management, illustrating the interdependency of Inclusive Cost Distribution, Resource Conservation, Smart Supply Management, and Climate Resilience. Together, these components form a cohesive strategy to balance economic efficiency, environmental stewardship, and resilience against climate challenges.

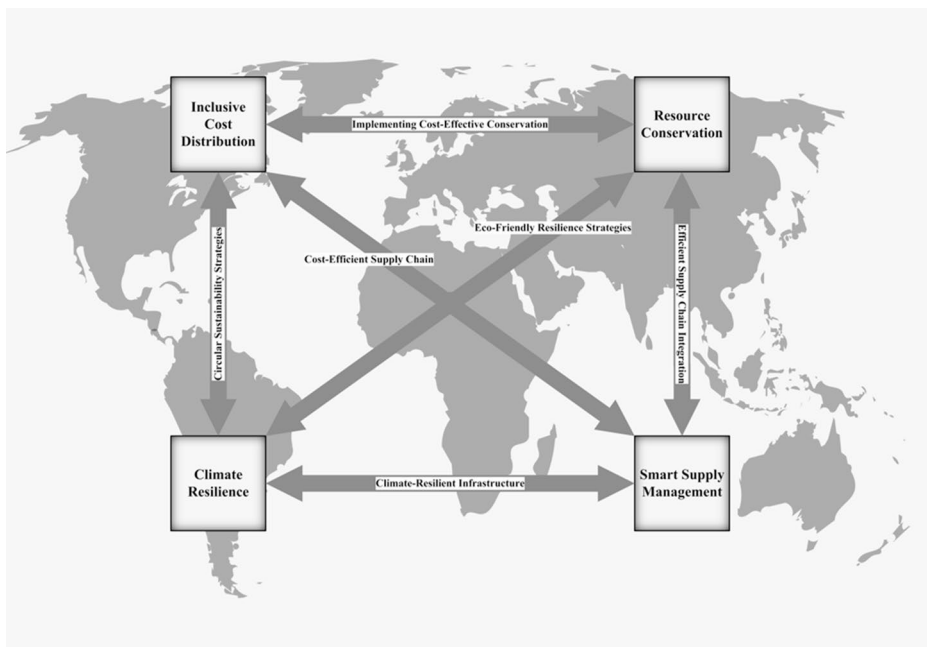


Fig. 4 Global framework for sustainable water resource management (reproduced from Leigh and Lee 2019)

3.2 Low Impact Development (LID)

Low Impact Development (LID) focuses on site-scale design to minimize stormwater management costs by utilizing natural drainage systems such as bioretention areas, green roofs, and swales, which are implemented near the source of runoff (Eckart et al. 2018). This approach contrasts with traditional end-of-catchment solutions like large-scale water treatment plants, as it aims to manage stormwater closer to where it originates (Fletcher et al. 2015). Bioretention areas or rain gardens are shallow, landscaped depressions that capture and treat stormwater with native vegetation that filters pollutants and promotes infiltration (Sharma and Malaviya 2021). Green roofs absorb rainfall, reduce runoff, and provide insulation, while swales, shallow vegetated channels, slow and convey stormwater, promoting infiltration and directing runoff to other LID features (Deksissa 2023). LID aligns with sustainable urban planning goals, addressing climate resilience and reducing the burden on municipal water management systems (Kumareswaran and Jayasinghe 2023).

3.3 Integrated Urban Water Management (IUWM)

Integrated Urban Water Management (IUWM) considers all components of the urban water cycle (Fig. 7) as interconnected (Mosleh and Negahban-Azar 2021). It addresses both ecological and anthropogenic water needs, ensuring sustainable water use based on local contexts. IUWM encompasses water supply, wastewater, stormwater, and their interactions, aiming to optimize the entire urban water system (Nieuwenhuis et al. 2021). IUWM and

Combined Sewer Overflows (CSOs) are closely related concepts in the field of urban water management (Manny 2023).

Kwon and Jung (2024) present a multiperiod optimization framework for urban drainage system (UDS) planning that addresses uncertainties and extreme climate conditions through scenario planning. Xu et al. (2024) discuss the application of Agent-Based Modeling (ABM) in water science, emphasizing its ability to simulate complex interactions at both the macroscale and microscale. Orgoványi and Karches (2024) focus on improving urban water utility networks through a GIS-based approach that aligns consumption data with network topology. Daneshgar et al. (2024) describe the development of a full-scale digital twin (DT) for the Eindhoven water resource recovery facility in the Netherlands, which enables real-time simulations, predictive maintenance, and scenario analysis. Montoya-Coronado et al. (2024) evaluate modeling practices for SUDS in mitigating Combined Sewer Overflows (CSOs), analyzing diverse approaches and performance indicators across 50 case studies. Park et al. (2024) examine the application of network analysis in urban flood research, with a focus on utilising green-grey infrastructure to mitigate urban pluvial floods.

Figure 5 represents the IUWM Index, structured into three concentric layers that collectively depict a comprehensive framework for urban water management. The innermost layer, labeled “IUWM Index,” serves as the core objective, highlighting the goal of integrated management across multiple domains. The second layer delineates the four main dimensions of the index: Governance, Social, Economic, and Environment. These dimensions provide a broad categorization of factors essential for a holistic approach to water management. The outermost layer comprises specific indicators within each dimension, offering detailed criteria to assess performance and guide improvements. The Economic dimension focuses on financial and resource management, including cost recovery and asset management, which are crucial for sustainable service delivery (Bleischwitz et al. 2017). Finally, the Environment dimension encompasses indicators such as stormwater management, wastewater management, and ecosystem health, emphasizing the need to maintain and protect natural resources (Wu et al. 2020).

3.4 Water Sensitive Urban Design (WSUD)

Emerging trends in digital tools, including digital twins, smart water systems, and AI-driven solutions, are increasingly vital to urban resilience, sustainability, and water management. These innovations, such as Digital Twins (DT) and Real-Time Control (RTC) for Nature-Based Solutions (NBS), enhance decision-making by offering real-time data analysis, predictive modeling, and enhanced system control, which directly address the challenges posed by climate change and urbanization. The “Industrial Internet of Water Things” (IIoWT) architecture leverages blockchain and digital twins to facilitate data standardization and secure interoperability within smart city water management (Mohammed et al. 2023).

Figure 6 illustrates various strategies for managing water supply, stormwater, and wastewater, organized into three primary sectors: Water Supply (green), Stormwater (blue), and Wastewater (orange). Each sector consists of three concentric core layers, i.e., Water Supply, Stormwater, and Wastewater. The middle layer elaborates on specific strategies within these categories, such as groundwater abstraction and water-saving devices for the water supply, rainwater harvesting devices and swales for stormwater management, and septic tanks and bioreactors for wastewater treatment. The outer layer indicates the scale or application level

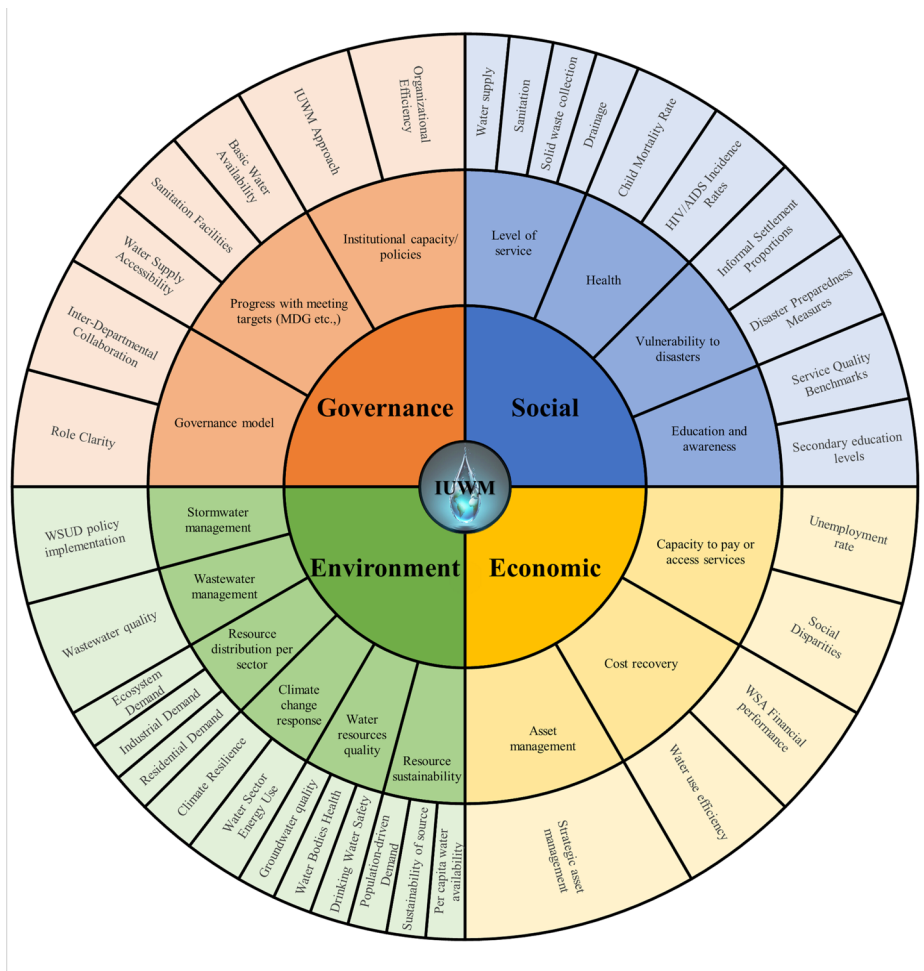


Fig. 5 An integrated urban water management approach covering governance, social, economic, and environmental dimensions

for each strategy, ranging from small-scale interventions to large-scale implementations. This structured approach provides a holistic understanding of water management, showcasing a spectrum of solutions tailored to different needs and scales.

Table 1 provides a comprehensive overview of diverse WSUD approaches and strategies, categorized by their application scale, benefits, challenges, and relevant parameters. It covers methods such as demand reduction, rain gardens, green roofs, rainwater tanks, and pervious pavement, each offering unique advantages, including reduced water stress, enhanced biodiversity, and flood mitigation. Table 1 also highlights the necessary tools/software for implementation and emphasizes the importance of community engagement for successful adoption.

The proposed strategies, SUDS, LID, and IUWM, contribute to the development of socio-ecological systems for sustainable urban water management. SUDS (Ellis and Lundy 2016)

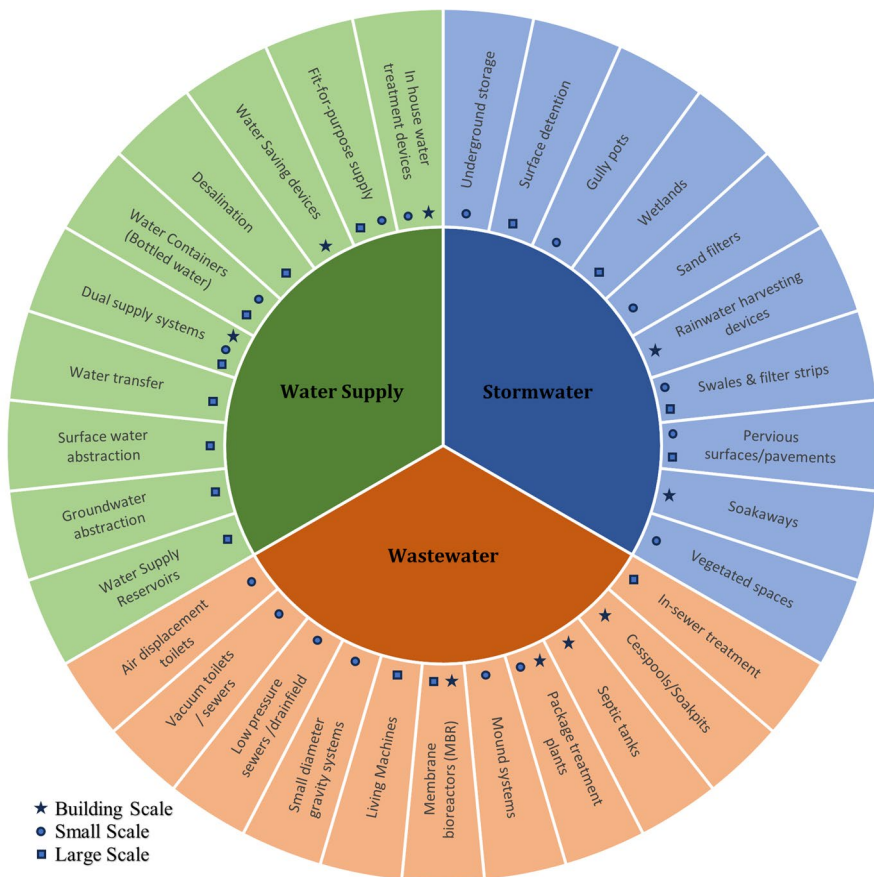


Fig. 6 An integrated approach to water supply, stormwater, and wastewater solutions across various scales

contribute to the restoration of infiltration and support biodiversity, thereby enhancing ecosystem services. LID (Fletcher et al. 2015; Sharma and Malaviya 2021) supports socio-ecological resilience by integrating green roofs, rain gardens, and permeable pavements into urban design, thereby reducing runoff and improving water quality while creating recreational and aesthetic benefits for communities. IUWM (Mosleh and Negahban-Azar 2021) provides the institutional and governance framework that links ecological and social dimensions, ensuring coordinated management of supply, stormwater, and wastewater. Together, these strategies embody socio-ecological resilience by addressing both natural ecosystem processes and human well-being. In addition, sociopolitical factors, including governance structures, institutional capacities, and political priorities, significantly influence the implementation of WSC (Koop et al. 2017; Rogers et al. 2020). For instance, regulatory frameworks and funding mechanisms determine whether decentralized solutions can be scaled effectively.

Table 1 Compilation of diverse water sensitive urban design (WSUD) approaches and strategies

S. No.	WSUD approaches	Scale	Benefits	Challenges	Parameters considered	Tools/Software	Community engagement	Remarks
1	Demand Reduction	City-wide, Municipal, Community	- Reduced water stress - Potential cost savings (utilities and consumers) - Reduced strain on water resources - Enhanced biodiversity - Flood mitigation potential - Urban heat island reduction (green roofs) - Aesthetic improvement and potential useable green space	- Promoting wide-spread behaviour change (adoption of water-efficient practices) - Sustaining public awareness campaigns for long-term impact - Cost of installation (including sourcing native plants) - Potential structural modifications (especially for intensive green roofs) - Long-term maintenance costs - Need for accessible knowledge & resources (design guides, workshops)	- Water consumption patterns - Historical and projected demand trends - Socioeconomic factors	- WaterGEMS, WaterCAD, SCADA Systems, Smart Metering Systems - Billing analysis tools	- Public awareness and education campaigns - Targeted outreach programs - Incentives for water-saving devices - Partnerships with schools and universities - Stakeholder involvement and education - Demonstration projects - Volunteer opportunities	Strategies and technologies aimed at reducing overall water consumption in urban areas through efficient use and conservation measures.
2	Rain Gardens, Green Roofs	Local, Neighbourhood, Site-specific	- Enhanced biodiversity - Flood mitigation potential - Urban heat island reduction (green roofs) - Aesthetic improvement and potential useable green space	- Cost of installation (including sourcing native plants) - Potential structural modifications (especially for intensive green roofs) - Long-term maintenance costs - Need for accessible knowledge & resources (design guides, workshops)	- Site suitability (soil infiltration, available space) - Local plant species (drought and flood tolerance) - Rainfall patterns (monsoon intensity and annual averages) - Local stormwater management regulations	GIS, SWMM, HydroCAD	Depressions in the landscape and vegetated roofs are designed to capture and filter stormwater runoff, promoting infiltration and reducing the risk of flooding. Designs can be adapted to a variety of site conditions, beautifying the city, and can be integrated with other WSUD features.	Depressions in the landscape and vegetated roofs are designed to capture and filter stormwater runoff, promoting infiltration and reducing the risk of flooding. Designs can be adapted to a variety of site conditions, beautifying the city, and can be integrated with other WSUD features.
3	Rainwater Tanks	Individual, Household, Commercial	- Water conservation - Potential cost savings (for users) - Reduced demand on potable water	- Initial installation costs - Maintenance requirements - Space constraints	- Rainfall patterns (including intensity and seasonality), - Storage capacity (based on intended uses), - Water quality (especially if intended for potable use)	RainHarvest Systems, DROP	- Community workshops and outreach - Hands-on training sessions on tank installation	Collection and storage of rainwater for later use, reducing demand on municipal water supplies and mitigating stormwater runoff. Offers potential water security during dry periods.
4	Pervious Pavement	Local Neighbourhood, Municipal/City-Wide	- Reduced runoff and flooding risk - Promotes groundwater recharge, supporting long-term water supply. - Potential water quality improvement	- Durability under heavy loads/climatic conditions - Maintenance requirements (regular cleaning needed) - Requires specialized installation knowledge	- Infiltration rates (soils and design layers), - Runoff patterns (including peak monsoon events) - Intended use (load-bearing needs)	PerviousPave (design features), SWMM (for advanced hydrologic modelling)	- Collaboration with local businesses for implementation on suitable sites - Explore potential incentives for participating businesses	Surfaces that allow water to pass through, reducing runoff and promoting groundwater recharge. Common materials include permeable concrete, asphalt, and pavers.

Table 1 (continued)

S. No.	WSUD approaches	Scale	Benefits	Challenges	Parameters considered	Tools/Software	Community engagement	Remarks
5	Urban Water Harvesting/Reuse	Household, Municipal, Industrial, Commercial	- Promotes Sustainable water use - Reducing strain on traditional sources & mitigating scarcity during droughts - Potential for cost savings (for users)	- Meeting target quality for end-use - Treatment standards - Regulatory compliance constraints - Public perception barriers	- Source Water quality variations & 'fit-for-purpose' targets based on intended reuse, - Assessment of existing Distribution infrastructure suitability (retrofitting needs vs. new system compatibility)	- Aquacycle (for assessment & sizing of reuse systems) - EPANET (for water quality and distribution network modelling in reuse schemes) - SCADA Systems (Real-time monitoring & control (SCADA Systems) for reuse scheme operation)	- Community participation in water reuse programs - Awareness campaigns (benefits of reuse) - Consultation of various stakeholder groups - Citizen science, volunteer initiatives	Urban water harvesting/reuse captures diverse water sources (rainwater, stormwater, greywater) for various uses, reducing strain on potable water while promoting sustainability and climate resilience. Systems range from household to city-wide scales.
6	Gross Pollutant Traps(GPT)	Local, Storm-water Drains, Watercourses	- Improved water quality - Protects aquatic habitats - Enhances amenity value of water bodies	- Frequent maintenance - Pollutant-specific effectiveness - Flow variability - Neglect causes odor & visual issues	- Target Pollutant types & size, - Load volume & flow regime - Regulatory compliance targets	- SWMM (catchment-level analysis), - GIS (site planning & operations)	- Public awareness campaigns - Ensure inclusive approach & outreach options - Targeted campaigns on pollution impact	Devices designed to capture and remove large debris and pollutants from stormwater runoff, preventing them from entering water bodies.
7	Bioretention Systems	Neighborhood, Industrial, Commercial	- Enhanced green infrastructure - Improves aesthetic & ecological functions - Potential for diverse water & placemaking benefits	- Space requirements (Land scarcity) - design complexity - Skill-based maintenance	- Infiltration rates, - Pollutant removal efficiency - Rainfall patterns impact sizing & operation	- HydroCAD (Flow calculations), - MUSIC (Pollutant modelling) - SWMM (Complex simulations)	- Collaboration with local environmental groups - Long-term stewardship model potential with active groups	Engineered systems that capture and treat stormwater through soil, vegetation, and filtration media, removing pollutants before releasing the water into the environment.
8	Swales and Buffer Strips	Local, Roadside, Watercourse Banks	- Erosion mitigation - Habitat enhancement - Potential for filtered runoff benefit & flood buffer - Aesthetic improvement	- Regular maintenance and Land use planning considerations - Long-term management needs dictate ownership planning - Vegetation lifespan	- Sedimentation rates, - Vegetative cover - Seasonal impact on vegetation density & performance	- GIS (Site suitability analysis), - SWMM (Flow & sediment modeling) - HydroCAD (Flow calculations & sizing)	- Involvement in community planning - Workshops aiding understanding of upkeep needs - Aesthetics feedback	Vegetated channels and strips are designed to convey, treat, and manage stormwater runoff. Buffer strips help filter pollutants and reduce erosion.

Table 1 (continued)

S. No.	WSUD approaches	Scale	Benefits	Challenges	Parameters considered	Tools/Software	Community engagement	Remarks
9	Sedimentation Basins	Local, Construction Sites, Industrial Areas	• Sediment reduction • Flood mitigation • Protect other assets or ecosystem health	• Land intensive Space requirements • downstream ecosystem sensitivity must be assessed • Maintenance of storage volume essential to function	• Influent dynamics must match design intent of basin • Sedimentation rates, Water quality • Seasonal flux in flows	• MIKE (Integrated Hydrology), HEC-RAS (Hydraulic Modeling)	• Collaboration with local authorities (permits, maintenance) • Community consultation (workshops, surveys) • Landscape integration feedback	Structures designed to slow down water flow, allowing sediment to settle before water is released, and reducing sedimentation in downstream areas. Can include baffles or vegetation to enhance trapping efficiency.
10	Constructed Wetlands	Municipal, Industrial, Restoration Projects	• Improves water quality (filtering pollutants like sediment, nutrients, metals, & pathogens) • Creates diverse wildlife habitat to enhance biodiversity • Mitigates flood risk by capturing & slowing runoff	• Long establishment period for full functionality • land use constraints • Requires specialised design expertise for long-term success • Requires maintenance (vegetation, hydraulics) • Perception challenges (mosquitoes, odors) in urban settings.	• Pollutant removal efficiency, • Habitat design for biodiversity • Hydraulic design (flow, retention) • Site factors (landscape, soils)	• Wetbud (Design Support), SWMM/MUSIC (Hydrologic Modeling), HEC-RAS (Hydraulic Analysis)	• Engagement with educational institutions • Citizen science initiatives • Educational workshops and site visits • Research collaboration & Internship opportunities	Constructed wetlands are versatile, engineered systems that mimic natural wetland processes to treat various water sources, offering flood control, habitat creation, and cost-effective solutions adaptable to diverse climates.
11	Wastewater Management	Municipal, Industrial, Large-scale decentralized Networks	• Water reclamation (irrigation, other uses) • Nutrient recovery (fertilizer) • Energy generation (biogas, heat) • Pollution control (reduces aquatic impacts) • Ecosystem protection • Potential socioeconomic benefits	• High Construction and land acquisition costs • Skilled labour and energy demands • Treatment complexity for specific pollutants • Maintenance burden • Regulatory compliance	• Target pollutant removal efficiency (BOD, suspended solids, heavy metals) and specific types (pathogens, specific heavy metals, emerging contaminants), • Nutrient forms and levels • Water quality impacts on receiving environment	• BioWin (biology-based treatment process modeling), SewerCAD (design and analysis of wastewater infrastructure), SCADA Systems (management, data monitoring, and control systems)	• Collaboration with industry stakeholders (for specialized treatment needs) • Include regulator input for compliance and quality • Seek community perspectives on impacts & benefits	Comprehensive management of wastewater, including treatment and recycling efforts to minimize environmental impact. Transforms wastewater into valuable resources – reclaimed water for diverse uses, nutrients for fertilizer, and potentially energy (biogas generation). Minimizes negative environmental impacts.

Table 1 (continued)

S. No.	WSUD approaches	Scale	Benefits	Challenges	Parameters considered	Tools/Software	Community engagement	Remarks
12	Siphonic Roofwater Systems	Commercial, Industrial, Large Rooftop Areas, Multi-story structures	• Rapid rainwater removal • Minimizes roof ponding & loads • Optimizes storage capacity efficiency • Lessens flash flood risk in urban areas	• Professional installation and maintenance cost • Increases upfront & maintenance cost • May discourage owners lacking technical knowledge	• Rainfall patterns • Rooftop configuration • Drainage capacity • Overflow prevention • Downstream storage (sizing & interface)	• FlowMaster (sizing and design) • Hydraflow (hydraulic modeling) • CFD packages for fine-tuning	• Technical Workshops for design professionals • Stakeholder engagement (architects and developers) • Benefits promotion • Public awareness	Siphonic roofwater systems offer an innovative solution for rapid, high-capacity rainwater removal, ensuring efficiency, preventing overflow, and optimizing space, particularly in dense urban environments.

4 Urban Water Cycle and its Management

Urbanization adversely affects the hydrological functions of catchments (Burns et al. 2012; Fletcher et al. 2013; Guan et al. 2015b; Salvatore et al. 2015) by altering the natural hydrological cycle. This alteration is primarily due to the introduction of impervious surfaces and the creation of artificial flow paths, resulting in decreased infiltration and reduced water storage capacity (Fletcher et al. 2013; Golden & Hoghooghi, 2018; Vogel et al. 2015). The urban water cycle (Fig. 4) involves inflows from both precipitation and imported water, which traverse the system, and the outflows include evapotranspiration, stormwater, and wastewater. The status of water storage is crucial for calculating changes in storage within the system.

The components of the entire urban water cycle that must be considered in an integrated biophysical assessment are depicted in Fig. 7. A suitable technique for evaluating the biophysical performance of the urban water system is to model water quantity and quality (Veerkamp et al. 2023). According to this approach, cities are considered as smaller catchments within larger catchments that supply water and receive stormwater and wastewater. These models incorporate water quality constituents (contaminants and pathogens), routing and hydrological processes, water extraction and allocation processes. Furthermore, they consider the feedback flows resulting from the usage of reclaimed water (Stormwater harvesting, recycling and greywater uses). Figure 7 presents a comprehensive overview of an urban water management system, detailing the interconnected processes of water sourcing, treatment, recycling, and usage. Starting with the water treatment plant, potable water is distributed to meet urban demand, supported by a service reservoir that ensures a consistent supply. Urban areas utilize this water for various purposes, including greywater and roofwater harvesting for irrigation and non-potable uses (Zhang et al. 2009). Stormwater is captured and treated to mitigate runoff and supplement the water supply. Decentralized Wastewater Treatment Plants (DWWTPs) recycle wastewater locally, whereas centralized wastewater treatment plants process the bulk of urban wastewater, producing treated effluent for indirect potable use or safe discharge. Additionally, a desalination plant provides

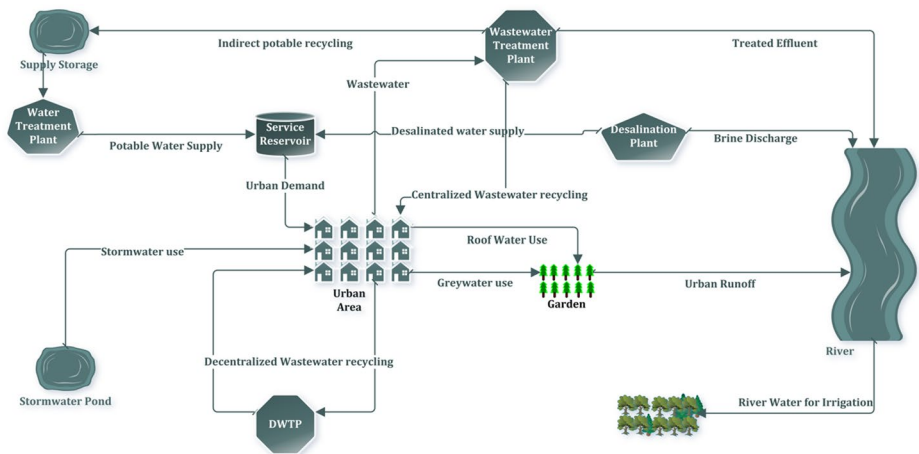


Fig. 7 Elements of the complete urban water cycle

freshwater by treating seawater, with brine discharge managed appropriately. Urban runoff can be controlled to prevent pollution and flooding, with treated effluent used to sustain river water levels for agricultural irrigation. This conceptualized integrated system (Fig. 7) highlights sustainable water management practices, optimizing resource use, and reducing dependency on external water sources.

Figure 8 highlights the multifaceted challenges that urban water systems face, including ageing infrastructure, uneven distribution, water quality concerns, and the effects of climate change (Stoler et al. 2022). These issues result in significant consequences, including economic burdens, public health risks, environmental degradation, and social inequities (Wakhungu et al. 2021). The proposed solutions emphasize upgrading infrastructure, implementing advanced monitoring systems, and adopting smart water technologies to enhance efficiency and reliability (Rojek and Studzinski 2019; Paudel et al. 2022). Additionally, it advocates for decentralized water management, integrating water considerations into urban planning, and building climate-resilient infrastructure (Pokhrel et al. 2022). Public awareness campaigns and community engagement initiatives are crucial for fostering a culture of water conservation (Antwi et al. 2022). To implement these solutions, we suggest a series of strategic pathways, including collaborative stakeholder engagement, data-driven decision-making, and continuous monitoring and evaluation. Emphasizing the importance of research, innovation, capacity building, and international collaboration, the approach is holistic and interdisciplinary, aiming to create sustainable, resilient urban water management systems that can adapt to future challenges and ensure equitable access to water resources.

In addition, Fig. 8 highlights critical issues in the urban stormwater management system, including inadequate capacity, poor maintenance, obsolete infrastructure, a lack of integration with urban planning, and the effects of climate change (Hou et al. 2020). These problems lead to severe impacts, including frequent flooding, property damage, traffic disruptions, public safety concerns, environmental degradation, infrastructure vulnerability, health risks, and community disruption (Freed and Zimmerman 2022). To address these challenges, the vision for solutions encompasses integrated urban planning, the implementation of green infrastructure, and enhanced community engagement and education, with an emphasis on climate-resilient design, smart technology integration, cross-agency collaboration, regular maintenance protocols, and public-private partnerships (Zuniga-Teran et al. 2020). The pathways to achieving these solutions involve detailed hydrological assessments, GIS mapping and data visualization, community surveys, and feedback mechanisms (Rodrigues et al. 2022). Additional strategies include stormwater modelling, water quality analysis, cost-benefit analysis, regulatory compliance reviews, public awareness campaigns, and the implementation of green infrastructure plans. Moreover, it highlights the importance of emergency preparedness training to enhance the resilience of urban stormwater drainage systems against future challenges and ensure the safety and well-being of the community.

Moreover, Fig. 8 offers a detailed framework for wastewater management analysis, divided into four interconnected sections: Existing Problems, Impact, Vision (Solution), and Pathways. Existing Problems highlight significant issues, including outdated treatment facilities, insufficient sewer system capacity, challenges in contaminant identification, incomplete nutrient removal, and limited green infrastructure (Capodaglio 2021). It also highlights the issues with industrial discharge pre-treatment, urban runoff management,

	Existing Problems	Impact	Vision (Solution)	Pathways
Water Distribution Analysis	• Aging Infrastructure • Uneven Distribution and Pressure • Leakage and Non-Revenue Water • Water Quality Concerns • Inadequate Monitoring and Management • Climate Change Impacts • Population Growth • Urban Planning Challenges • Regulatory Compliance • Limited Community Awareness	• Economic Burden • Public Health Risks • Environmental Impact • Disruption of Services • Compromised Infrastructure Integrity • Increased Operating Costs • Legal and Regulatory Consequences • Social Inequities • Strain on Emergency Response • Degraded Quality of Life	• Infrastructure Upgrades • Advanced Monitoring Systems • Smart Water Technologies • Decentralized Water Management • Sustainable Urban Planning • Climate-Resilient Infrastructure • Water Conservation Programs • Public Awareness Campaigns • Community Engagement Initiatives • Regulatory Compliance Measures	• Collaborative Stakeholder Engagement • Data-Driven Decision Making • Research and Innovation Initiatives • Capacity Building and Training • Flexible and Adaptive Strategies • Integrated Water Resource Management • Continuous Monitoring and Evaluation • Policy Advocacy and Reform • International Collaboration • Holistic and Interdisciplinary Approaches
Storm Water Drainage Analysis	• Inadequate Capacity • Poor Maintenance • Obsolete Infrastructure • Lack of Integration with Urban Planning • Insufficient Grading and Sloping • Lack of Sustainable Practices • Urbanization Effects • Climate Change Impact	• Flooding Occurrence • Property Damage • Traffic Disruptions • Public Safety Concerns • Environmental Degradation • Infrastructure Vulnerability • Health Risks • Community Disruption	• Integrated Urban Planning • Modernized Infrastructure • Green Infrastructure Implementation • Community Engagement and Education • Climate-Resilient Design • Smart Technology Integration • Cross-Agency Collaboration • Regular Maintenance Protocols • Public-Private Partnerships • Resilient Emergency Response Plans	• Hydrological Assessment • GIS Mapping and Data Visualization • Community Surveys and Feedback • Stormwater Modeling • Water Quality Analysis • Cost-Benefit Analysis • Regulatory Compliance Review • Public Awareness Campaigns • Green Infrastructure Implementation Plan • Emergency Preparedness Training
Waste Water Management Analysis	• Outdated Treatment Facilities • Inadequate Sewer System Capacity • Contaminant Identification Challenges • Incomplete Nutrient Removal • Limited Implementation of Green Infrastructure • Insufficient Pre-treatment of Industrial Discharges • Urban Runoff Management Challenges • Ineffective Sludge Management • Limited Application of Advanced Treatment Technologies • Insufficient Monitoring of Non-Point Source Pollution	• Water Quality Degradation • Public Health Risks • Ecosystem Disturbance • Aquifer Contamination • Impact on Agriculture • Adverse Effects on Industries • Tourism and Recreational Impacts • Increased Treatment Costs • Legal and Regulatory Consequences • Social Equity Concerns	• Infrastructure Modernization • Smart Monitoring Systems • Community Education and Engagement • Stringent Industrial Regulations • Promotion of Sustainable Technologies • Climate-Resilient Infrastructure • Regulatory Framework Enhancement • Increased Budget Allocation • Integrated Governance and Collaboration • Research and Innovation Support	• Integrated Water Resource Management • Circular Economy Principles • Decentralized Wastewater Treatment • Nature-Based Solutions • Innovation Hubs for Research and Development • Public-Private Partnerships • Capacity Building and Training Programs • International Collaboration and Knowledge Exchange • Public-Private Innovation Challenges • Continuous Monitoring and Adaptive Management

Fig. 8 Urban water system analysis (water distribution, stormwater and wastewater aspects)

ineffective sludge management, limited access to advanced treatment technologies, and inadequate monitoring of non-point source pollution. The Impact section outlines the consequences of these problems, including water quality degradation, public health risks, ecosystem disturbances, aquifer contamination, agricultural impacts, adverse industrial effects, and increased costs, among others (Kesari et al. 2021). The vision proposes comprehensive strategies, including infrastructure modernization, smart monitoring systems, community education, stringent industrial regulations, sustainable technologies, climate-resilient infrastructure, regulatory enhancements, increased budget allocation, and integrated governance (Wheeler 2021). Finally, the pathways section proposes methods for implementing these solutions, including integrated water resource management, circular economy principles, decentralized wastewater treatment, nature-based solutions, innovation hubs, public-private partnerships, capacity building, international collaboration, public-private innovation challenges, and continuous monitoring and adaptive management (Lebu et al. 2024). This detailed approach underscores the necessity of multifaceted and collaborative efforts to effectively manage wastewater.

The comprehensive investigation of existing challenges in urban water management is presented in the supplementary material. Furthermore, a detailed account of the evolution of urban water systems has been provided to illustrate the progressive development of frameworks and management approaches over time.

5 Selected Case Studies

To explore the applicability of the concept of a water-sensitive city, we have reviewed various case studies. A few of these are discussed below to illustrate the concept.

5.1 United Kingdom: Starting from the Source with One Bird and Two Purposes

The British government has been addressing the increasingly severe water shortage problem and the need to improve municipal drainage capacity in large cities, such as London, by encouraging residents, communities, and businesses to install rainwater collection systems (White 2010). This approach aimed to solve both issues at their source. Most homes have storage tanks that hold between 1,000 and 7,500 L of rainwater, which is collected from roofs, filtered, and stored in underground tanks (Rahman 2021). The UK is also promoting the use of rainwater in large municipal and commercial buildings. For example, the London Olympic Park recycles rainwater and wastewater, providing all the irrigation water for the 225-hectare park (Grant 2012). Additionally, the park supplies recycled rainwater and reclaimed water to nearby residents.

5.2 France: Different Forms of the Ascension Cycle

France has unique approaches to managing drainage, flood prevention, and rainwater recycling in its major cities (Chocat et al. 2001). Paris, as the capital, boasts a water system that's seen as a global model (Goodell 2018). Designed in 1852 by Haussmann, the system collects surface water through underground pipes, aiming to keep the Seine River clean (Gordon 2006). In Lyon, another notable French city, the focus is on natural water recycling, using the natural slope of its land to guide rainwater into specially designed canals in community areas.

5.3 Germany: Efficient Water Harvesting Balances Ecology

Germany has successfully built "sponge cities" by using advanced underground pipes, rainwater technologies, and green spaces (Nguyen et al. 2019). The city (Berlin and Siemensstadt Square) introduced a system called the "depression-infiltration canal system," which links low-lying areas and canals to drainage pipes with holes. This decentralised system temporarily stores rainwater in grasslands and canals, thereby easing pressure on urban drains (Wu et al. 2019). Over the years, Germany has refined rainwater technologies. They focus on three primary methods, including collecting rainwater from roofs, intercepting rainwater for infiltration, and using rainwater in ecological communities. They're also experimenting with ideas like large city parks and encouraging citizens to create "green roofs", which may retain up to 60% of future rainfall (Zölch et al. 2017).

5.4 Singapore: Strict Standards for Good Guidance

Singapore, a tropical island nation with abundant rainfall, has successfully managed to prevent urban waterlogging despite increasing annual rainfall over the past three decades. Singapore plans its urban drainage system meticulously before constructing any buildings

(Chan et al. 2018). It includes designing both underground and surface drainage systems with dedicated drainage canals integrated into every building, sidewalk, and road. They have implemented a comprehensive urban rainwater collection system. Rainwater collected from the city is channelled into strategically located 17 large reservoirs. Singapore's Public Utilities Board ensures that new constructions incorporate heightened waterproofing measures, thereby reducing the risk of water damage.

5.5 Los Angeles, USA: Strengthen Design and Speed Up Reconstruction

In most US cities, stormwater management encompasses both structural solutions, such as detention basins and green roofs, and non-structural practices, including public education and ordinances (National Research Council 2009). Additionally, traditional water management involves collecting and storing rainwater in suburban areas, channelling it to urban centres through canals, and conveying sewage through underground drainage systems. They're exploring how to capture rainwater for drinking and irrigation, imagining cities that act like sponges to absorb rain effectively. Architects and planners, such as those working on the "sponge city" concept in Le Perle, France, believe that future trends will blur the lines between cities and water (Novotny 2020).

In view of the case studies presented above, the WSC concepts used by the cities are governed by: (1) Geographical and Climatic Factors, (2) Technological Approaches and Infrastructure, (3) Economic Context and Resource Allocation, (4) Social and Cultural Engagement, and (5) Environmental Impact and Resilience.

6 Conclusion

This review has explored the principles, methodologies, and global experiences, emphasizing WSC's critical role in securing urban sustainability and resilience. At its core, WSC recognizes the interconnectedness of water supply, stormwater, and wastewater systems. This results in optimizing water resource utilization, minimizing pollution, and enhancing adaptability to the challenges of climate change. The increasing adoption of SUDS, LID, and IUWM principles is a clear sign of this paradigm shift. This collaborative approach ensures that water management strategies are comprehensive, encompassing social, economic, and environmental dimensions. The WSC framework also emphasizes the need for "fit-for-purpose" solutions, tailored to the specific context of each city. This means considering unique factors, such as climate, topography, existing infrastructure, and social dynamics, rather than relying on one-size-fits-all approaches.

Furthermore, WSCs prioritize decentralized water management, empowering local communities to become active participants in managing water resources. The suggested nature-based solutions not only manage stormwater runoff and improve water quality but also create aesthetically pleasing public spaces. Effective communication, public education, and stakeholder engagement are vital for building consensus, promoting responsible water use, and ensuring the long-term success of WSC initiatives.

The case studies presented in this review illustrate the diverse approaches employed by various countries to achieve WSC goals. While each city has unique challenges and

resources, the common theme that emerges is a commitment to sustainability, resilience, and community engagement.

Despite significant progress, challenges persist, including securing funding and supportive policy frameworks, the application of advanced technology in water treatment, the development of monitoring systems and data analysis, and ensuring equitable access to clean water and sanitation services. By fostering integrated approaches, promoting fit-for-purpose solutions, and prioritizing community engagement, cities can move towards WSC. The journey to WSCs is not a destination but an ongoing process of innovation, adaptation, and collaborative action.

7 Key Recommendations for a Sustainable Future

The pursuit of Water Sensitive Cities (WSCs) requires a multifaceted approach, integrating technological innovation, policy reform, community engagement, and ongoing research. Key recommendations to advance WSC implementation globally, with a particular focus on rapidly urbanizing regions, are presented below.

7.1 Policy and Governance

To promote holistic water management, it is essential to develop integrated policies that encompass water supply, stormwater, and wastewater management, recognizing their interconnectedness. Decentralized water management should be prioritized, encouraging small-scale interventions like rainwater harvesting and greywater reuse to empower communities and reduce reliance on centralized systems. Financial incentives and regulatory frameworks must be developed to promote the adoption of green infrastructure in both new and existing urban spaces. Strengthening regulatory oversight is crucial, with robust regulations for industrial discharge, effective pre-treatment, and monitoring of non-point source pollution to safeguard water quality. Additionally, fostering inter-departmental collaboration among water supply, wastewater, and stormwater agencies will facilitate comprehensive water management planning. Investing in capacity building through training programs for urban water management professionals is also vital, as it equips them with the knowledge and skills necessary for implementing WSC principles.

7.2 Technology and Infrastructure

Advancing water treatment technologies is crucial, with investments in research and development focusing on cost-effective and efficient methods, particularly for decentralized systems and greywater reuse. Integrating smart technologies, such as sensors, data analytics, and decision support systems, can optimize resource utilization and detect leaks efficiently by enabling real-time monitoring of water systems. Embracing climate-resilient infrastructure involves designing and retrofitting urban areas to withstand extreme weather events, including heavy rainfall and droughts. Innovative flood management strategies should move beyond traditional control measures, incorporating structural solutions, early warning systems, public education, and community preparedness. Additionally, promoting water-

efficient fixtures and appliances in households and commercial buildings can significantly reduce overall water consumption, contributing to sustainable water management.

7.3 Community Engagement and Awareness

Engaging communities in water management planning is essential, as it involves residents in the development and implementation of WSC initiatives to foster a sense of ownership and promote local solutions. Public awareness campaigns should be conducted to educate communities about responsible water use, the challenges of water scarcity, and the benefits of WSC practices, thereby promoting water conservation education. Building public trust and transparency requires providing clear and accessible information about water management policies, infrastructure projects, and data collection efforts, ensuring public trust and accountability in the management of water resources.

7.4 Research and Development

Conducting comprehensive urban water metabolism analysis is crucial, involving the development of robust models to analyze urban water flows, identify key management challenges, and evaluate the effectiveness of Water Sensitive Cities (WSC) interventions. Emphasizing adaptive management, ongoing monitoring and evaluation programs should be implemented to assess the impact of WSC strategies, identifying areas for improvement and adapting to changing conditions. Innovative financing mechanisms need to be explored, investigating funding sources and investment strategies to support WSC implementation, particularly in developing countries. Additionally, studying the long-term performance of WSC solutions is essential, with research focusing on the sustainability and resilience of green infrastructure, decentralized systems, and other WSC interventions.

By focusing on these recommendations, cities can successfully transition towards Water Sensitive Cities, ensuring sustainable, resilient, and equitable access to water resources for present and future generations. This collaborative approach requires a commitment to ongoing learning, adaptation, and a shared vision for a water-secure future.

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Declarations

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